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I'm a biomedical engineer turned molecular biologist turned data scientist who has spent my career at the intersection of discovery and translation in cancer research. **My work has always centered on connecting the technical with the human, building bridges between wet-lab scientists, clinicians, and data and technology professionals** to make biomedical research more reproducible, scalable, and impactful.

I've spent the last decade **identifying data needs, developing internal and external data partnerships, integrating clinical and molecular oncology data, and establishing governance for the appropriate use of sensitive healthcare data.** In leading multidisciplinary teams spanning data governance, engineering, analytics, and data science, I've advised scientific, clinical, technical, legal, compliance, and institutional leaders on selecting and using data to support oncology research and decision-making.

Core Strengths & Expertise

- **Oncology Data Strategy & Partnerships:** I have developed data and technology strategies to connect comprehensive, multimodal oncology data with priority scientific and organizational use cases in a Comprehensive Cancer Center setting. I've fostered collaborations across healthcare institutions, research organizations, technology partners, and national consortia.
 - **Comprehensive Data Products & AI Readiness:** I have built and led matrixed, multidisciplinary teams that acquire, harmonize, model, and operationalize complex healthcare and molecular data, enabling its use across analytics, clinical trials, laboratory research, and AI. This has included developing the infrastructure, governance, and AI policies needed to support responsible AI adoption while protecting patient privacy, ethical use, and data security.
 - **Data Strategy, Governance & Leadership:** I have developed and executed translational data strategy and governance processes spanning clinical and research domains. Implementing these programs has required building and leading cross-functional committees, governance structures, working groups, and project teams; managing change and communications; and mentoring staff across technical and scientific disciplines.
 - **Specific Areas of Expertise:**
 - **Oncology data:** EHR/clinical, OMOP, genomic and molecular profiling, laboratory data, pathology, registry data, longitudinal outcomes, multimodal datasets
 - **Data partnerships:** healthcare institutions, external data providers, technology partners, national consortia, data-use agreements, inter-institutional sharing
 - **Governance:** HIPAA, IRB, privacy, security, data stewardship, NIH genomic data, contractual restrictions, international data sharing
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Experience

Associate Vice President & Deputy Chief Data Officer (*promoted from Senior Staff Scientist, Data Science Lab*) | 2022–Present | Fred Hutchinson Cancer Center — Seattle, WA

- **Lead strategy and operations for Fred Hutch's translational oncology data portfolio**, spanning clinical, molecular, genomic, research, and external data assets and the governance, engineering, analytics, and data science capabilities required to make those data usable. Enabled Fred Hutch's participation in the Cancer AI Alliance, a national oncology collaboration using federated data and learning approaches.
- **Develop and oversee oncology data partnerships and data-sharing relationships** with healthcare organizations, clinical laboratories, research collaborators, national consortia, and external data providers, from identification of data needs and evaluation of available sources through contracting and legal review, governance, integration, quality review, and ongoing use.
- **Lead evaluation, acquisition, and fit-for-purpose use of internal and external oncology data**, working with institutional and consortium clinical laboratories and external providers including Tempus, Caris, Foundation Medicine, and others to assess data availability, limitations, quality, and appropriate use for precision oncology, translational research, and analytics.
- **Established institutional data governance and stewardship frameworks** governing data access, inter-institutional sharing, data-use agreements, regulated human-subjects data, NIH genomic-data requirements, HIPAA, and international data-sharing requirements.
- **Led development of multimodal oncology data products** linking longitudinal clinical data with genomic and molecular profiling, laboratory data, pathology and other research data, including internally generated and externally sourced datasets; established common models, provenance, access controls, and governance to support precision oncology and downstream analytics.
- **Built and led a multidisciplinary data organization** spanning data governance, engineering, analytics, data science, and training; established portfolio strategy, operating models, staffing plans, and multimillion-dollar budgets and hired 29 team members in 24 months.
- **Architected cloud-based data infrastructure** (Databricks, GitHub, AWS, plus open source data visualization applications) to enable scalable data products and ethical AI applications supporting precision oncology and translational research.

Senior Staff Scientist, Translational Genomics & Bioinformatics | 2016–2022 | Fred Hutchinson Cancer Center — Seattle, WA

- **Led a multidisciplinary translational oncology research program** supported by multimillion-dollar NIH funding, integrating microfluidics, genomics, clinical data, and computational methods to study tumor evolution and treatment resistance across multiple cancer types.
- **Translated scientific and clinical questions into data requirements** and analytical approaches, working with investigators, laboratory teams, computational scientists, and collaborators to identify appropriate datasets, understand data limitations, and design reproducible analyses.
- **Created and managed the Translational Genomics Repository**, a cloud-based data resource containing 1,500+ translational research datasets across prostate, lung, leukemia, and breast

cancer programs. Established metadata, ontology harmonization, indexing, provenance, and reusable data organization practices across collaborative studies.

- **Integrated complex genomic, clinical, experimental, and laboratory metadata** across projects, resolving inconsistencies in data definitions and structure to support cross-study analysis and reuse.
- **Developed reproducible data pipelines and analytical tools** for exome, targeted sequencing, single-cell RNA sequencing, and imaging data, with emphasis on data quality, traceability, and potential downstream clinical and CLIA applications.
- **Received a K01 award from the NIH Big Data to Knowledge Initiative** to advance development of pipelines for image analysis in R, custom analysis workflows for novel exome, targeted DNA and single cell RNA sequencing genomic datasets and data integration with a focus on reproducibility to enable potential future CLIA assay validation. (K01-ES026842)

Data Strategist | 2016–2018 | Fred Hutchinson Cancer Center — Seattle, WA

- **Advised the Chief Data Officer on institutional strategy for translational and clinical research data management.** Shaped early policy and infrastructure recommendations that informed later governance and data modernization efforts.
- **Advised investigators on acquisition, use, and sharing of sensitive and externally sourced data,** including advising on data policy and IRB oversight of sensitive and/or restricted data including those falling under GDPR, HIPAA and other regulatory or contractual requirements.
- **Partnered with legal, compliance, IRB, security, and technical teams** to interpret HIPAA, GDPR, institutional policy, and contractual requirements and translate them into practical data access and stewardship processes.
- **Helped investigators assess whether available data were appropriate for proposed research questions,** including consideration of consent, permitted use, data provenance, access restrictions, and downstream sharing requirements.

Staff Scientist | 2013–2016 | Fred Hutchinson Cancer Center — Seattle, WA

- **Led laboratory and computational oncology research** integrating microfluidic single-cell assays, sequencing, clinical samples, and large-scale data analysis to study clonal evolution in leukemia.
- **Managed multidisciplinary collaborations spanning clinicians, laboratory scientists, computational researchers, and external partners,** aligning experimental design, specimen and data requirements, analysis plans, and scientific objectives.
- **Developed and applied methods for integrating heterogeneous biological and genomic data,** including single-cell and sequencing datasets, with emphasis on reproducibility, data quality, and interpretable results.

- **Secured and managed multimillion-dollar federally funded and collaborative research programs**, including NIH R01, K award, pilot, SBIR, institutional, and privately funded projects.
 - **Mentored clinical, laboratory, and computational staff** and drove adoption of R, Python, GitHub, containerized workflows, and reproducible research practices.
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Education

Academic Publications (available via [NCBI MyBibliography](#))

Doctor of Philosophy in Biomedical Engineering - 2008

University of Wisconsin (Madison, WI);

Concentration: Microfluidic System Design, Biological Assays, Cancer Biology

Master of Science in Biomedical Engineering - 2005

University of Wisconsin (Madison, WI);

Concentration: Modeling of Dynamic Biological Systems, Biopolymer/Polymer Science

Bachelor of Science in Mechanical Engineering - 2003

University of Illinois (Urbana, IL);

Concentration: Flow Cytometry, Biofluid Dynamics, Microhemodynamics

Bachelor of Science in Biology - 2003

University of Illinois (Urbana, IL);

Concentration: Bioengineering and Biophysics
